Please take five minutes to complete a course evaluation. You should get an e-mail with a link.

Problem 1. This problem will walk us through defining $\mathbb{Q}$ using an equivalence relation. Given $(a, b), (c, d) \in \mathbb{Z} \times (\mathbb{Z}\backslash\{0\})$, we say that

$$(a, b) \sim (c, d) \text{ if and only if } ad = bc.$$

For example $(1, 2) \sim (3, 6)$ but $(3, 4) \not\sim (9, 4)$.

1. Show that $\sim$ is an equivalence relation.

   Solution. We will show that $\sim$ is reflexive, symmetric, and transitive.

   Reflexive. Let $(a, b) \in \mathbb{Z} \times (\mathbb{Z}\backslash\{0\})$. By commutativity, $ab = ba$, which means that $(a, b) \sim (b, a)$. Thus $\sim$ is reflexive.

   Symmetric. Let $(a, b), (c, d) \in \mathbb{Z} \times (\mathbb{Z}\backslash\{0\})$ and assume that $(a, b) \sim (c, d)$. Thus $ad = bc$, so $cb = da$ (commutativity again), and thus $(c, d) \sim (a, b)$ by definition. Therefore $\sim$ is symmetric.

   Transitive. Let $(a, b), (c, d), (e, f) \in \mathbb{Z} \times (\mathbb{Z}\backslash\{0\})$ and assume that $(a, b) \sim (c, d)$ and that $(c, d) \sim (e, f)$. This means that

   $$ad = bc \tag{1}$$

   and

   $$cf = de. \tag{2}$$

   Multiplying Equation (1) by e and Equation (2) by a, we see that $ade = bce$ and $acf = ade$. Thus $acf = bce$. If $c = 0$, then I claim that $a = e = 0$ (check this!), and so $(a, b) \sim (e, f)$ since $0 = 0$. If $c \neq 0$, then by cancellation, $af = be$, and thus $(a, b) \sim (e, f)$. Thus $\sim$ is transitive.

   This shows that $\sim$ is an equivalence relation, which is great news. □

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2. Given $x \in \mathbb{Z} \times (\mathbb{Z}\backslash\{0\})$, recall that **the equivalence class** of x is

   $$[x] = \{y \in \mathbb{Z} \times (\mathbb{Z}\backslash\{0\}) \mid x \sim y\}.$$

   For example, $(-2, -6) \in [(1, 3)]$ but $(0, 4) \notin [(15, 25)]$. We can define

   $$\mathbb{Q} = \{[x] \mid x \in \mathbb{Z} \times (\mathbb{Z}\backslash\{0\})\}.$$

   So $\mathbb{Q}$ is a set of equivalence classes. Why does it make sense to define $\mathbb{Q}$ in terms of equivalence classes (rather than just all pairs in $\mathbb{Z} \times (\mathbb{Z}\backslash\{0\})$)?

   Solution. This makes sense because the same rational number can be represented in many ways. If you haven't noticed by now, the key observation to make here is that the pair (a, b) represents the number $\frac{a}{b}$. Thinking about, for example, the rational number $\frac{1}{2}$. There are tons of ways to represent it as a pair: $(1, 2)$, $(-5, -10)$, $(3, 6)$, etc. Thus collecting *all* of the pairs that represent each rational number and treating the entire equivalence class as one number is a sensible thing to do. □

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3. Let $[(a, b)]$ and $[(c, d)]$ be equivalence classes as defined in Part (b). Given that we want these to represent rational numbers, tell me how we should define the following:

   $$[(a, b)] + [(c, d)] = \ ?$$
   $$[(a, b)] \cdot [(c, d)] = \ ?$$

Note: Each question mark should be replaced by some equivalence class. You do not have to formally prove your answer, but you should explain why it makes sense for $\mathbb{Q}$.

Solution. The correct answers here are

$$[(a,b)] + [(c,d)] = [(ad+bc, bd)]$$
$$[(a,b)] \cdot [(c,d)] = [(ac, bd)]$$

This is all I'm asking for this problem. However, we defining this operation, we should ask if it is **well-defined.** In this problem, this means that the operation works the same regardless of which representative we use for each equivalence class.

For example, we want $[(1,2)] + [(5,6)] = [(-3,-6)] + [(10,12)]$ and it's not clear from the definitions that these are the same. In general, we have to show that if we take two different representatives (a,b) and (a',b') of the same equivalence class that these operations work the same regardless of which representative we pick. We unfortunately don't have time to focus on this in our class, but it's something to be aware of. It comes up a lot in math, especially when we based definitions on equivalence classes. □

Problem 2. Assume that A and B are finite sets with $|A| = |B|$. Let $f\colon A \to B$ be a function. Prove that f is injective if and only if f is surjective.

Proof. Suppose that A and B are finite sets with $|A| = |B|$

[$\Rightarrow$] We prove the contrapositive statement. Suppose that $f\colon A \to B$ is not surjective. We want to show that it is not injective. Since f is not surjective, there exists some $b \in B \backslash f(A)$. Now we define a function $g\colon A \to B\backslash\{b\}$ by the rule $g(a) = f(a)$ for any $a \in A$. Note that g indeed lands in $B\backslash\{b\}$ because b is not in the image of f. Now, since $b \in B$, by a theorem proved in class we know that $|B\backslash\{b\}| = |B| - 1 < |B| = |A|$. Thus, by the Pigeonhole Principle, $g\colon A \to B\backslash\{b\}$ cannot be injective, say $g(a_1) = g(a_2)$ for some distinct a_1 and a_2 in A. But then also $f(a_1) = f(a_2)$, so f is not injective.

[$\Leftarrow$] Again, we prove the contrapositive statement. Assume $f\colon A \to B$ is not injective. Let $x \neq y$ in A such that $f(x) = f(y)$. Now we define a function $h\colon A\backslash\{x\} \to B$ by the rule $h(a) = f(a)$ for any $a \in A\backslash\{x\}$. Since $|A\backslash\{x\}| < |B|$, by the surjective Pigeonhole Principle, h can not be a surjection. Since $h(A\backslash\{x\}) = f(A)$ then f can not be surjective, as desired. □

Problem 3. Assume that A and B are finite sets with $|A| < |B|$. Prove that there is no surjective function $A \to B$.

Note: This is the "surjective analogue" of the Pigeonhole Principle that we did not prove in lecture.

Hint: Contrapositive. Assume $f\colon A \to B$ is surjective. Define $g\colon B \to A$ by choosing, for each $b \in B$ some $a_b \in f^{-1}(b)$. Show that g is injective

Proof. Suppose A and B are finite sets and $f\colon A \to B$. We prove the contrapositive of the statement.

Suppose that f is surjective. Define a function $g\colon B \to A$ as follows. For each $b \in B$, since f is surjective, there exists some $a \in A$ such that $f(a) = b$ (there may be more than one), that is, $f^{-1}(b) \neq \varnothing$. Choose any such $a \in f^{-1}(b)$ and let $g(b) = a$. We claim that g is an injection. In fact, if $b \neq b'$, since f is a function $f^{-1}(b) \cap f^{-1}(b') = \varnothing$ so $g(b) \neq g(b')$.

Now by the Pigeonhole Principle, since $g\colon B \to A$ is an injection we have $|B| \leqslant |A|$, which is the negation of $|A| < |B|$. □

Problem 4. Assume that a party has $n \geqslant 2$ attendees. Some people know each other and some people do not. Assuming that Person A knows Person B if and only if Person B knows Person A,

show that there must be two people at the party who know the same number of people. (A person is not considered to know himself/herself so everyone knows between 0 people (if they know nobody at all) and $n-1$ people (if they know everybody).)

Hint: Consider the case where there's some person who knows no one and then the case where everyone knows at least one person. Apply the Pigeonhole Principle to each.

Proof. Let P be the set of the n people at the party. As suggested in the hint, there are two cases.

Case 1. If everyone knows at least one person, then consider the function $f\colon P \to \{1, 2, \ldots, n-1\}$ defined by $f(x) =$ the number of people that x knows. Since $|P| = n$ and $|\{1, 2, \ldots, n-1\}| = n-1$, by the Pigeonhole Principle, f is not injective. So we have$f(x) = f(y)$ for some $x \neq y$. But then x and y know the same number of people, and in this case we are done.

Case 2. Suppose someone at the party does not know anyone. There are two ways to prove this case. One way to prove it is to consider a similar function in as in Case 1. Since someone does not know anyone, this means that there is nobody at the party who knows everyone. So the number of people that anyone at the party knows is in the set $\{1, 2, \ldots, n-2\}$ (since nobody knows $n-1$ people). You then apply the Pigeonhole Principle as before.

An alternative proof for Case 2 is this: Suppose someone at the party does not know anyone. If there is someone else who does not know anyone, then these first two people both know 0 people so we are done. If not, then among the remaining $n-1$ people, they all know at least one of the other $n-1$ people. So we have reduced to Case 1 (with $n-1$ people). So there are two people in the group of $n-1$ who know the same number of people. □

Problem 5. Prove that the following sets are countably infinite. In other words, prove that they are equinumerous to $\mathbb{N}$.

1. The set of integers F that are multiples of 5.

   Solution. We know that $\mathbb{N} \approx \mathbb{Z}$, so it suffices to give a bijection $\mathbb{Z} \to F$. Define $f\colon \mathbb{Z} \to F$ by $f(x) = 5x$. Then f is surjective, since every multiple n of 5 is of the form $5k$ for some integer k. Hence, $f(k) = n$.

   Further, f is injective since if $f(m) = f(n)$ then $5m = 5n$ whence $m = n$. □

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2. $A = \left\{\frac{1}{2^k} \mid k \in \mathbb{N}\right\}$.

   Solution. We directly give a bijection from $\mathbb{N}$ to the set A. Namely, define $f\colon \mathbb{N} \to A$ by $f(k) = 1/2^k$. This function is surjective by the definition of the set A. It is injective since if $f(m) = f(n)$ then $1/2^m = 1/2^n$ so $2^m = 2^n$ so $m = n$, by taking $\log_2$, which is an injective function. □

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3. $B = \{n \in \mathbb{Z} \mid n \geqslant -30\}$.

   Solution. We give a bijection from $\mathbb{N}$ to the set B. Namely, define $f\colon \mathbb{N} \to B$ by $f(k) = k - 31$. Then f is surjective, since if $n \in B$ then $n + 31 \geqslant 1$ so $n + 31 \in \mathbb{N}$ and $f(n + 31) = n$.

   Further, f is injective since if $f(m) = f(n)$ then $m - 31 = n - 31$ so $m = n$. □
